Proposal for a new course: Department of Physics

1. **Course Number**: 6XX
2. **Course Title**: Function Spaces for the Formalism of Quantum Mechanics
3. **Number of Lectures Per Week and Total Number of Classes**: 3 Lectures [Tutorial:0 ,Laboratory:0 ,Additional Hours:0], 39
4. **Credit**: 09
5. **Duration of Course**: Full Semester
6. **Proposing Department and Type of Course**: Physics, Departmental Elective
7. **Other Members Interested in Teaching The Course**: Diptarka Das , Nilay Kundu , Apratim Kaviraj
8. **Proposing Instructor**: Kaushik Bhattacharya
9. **Course Outline**

   (a) **Introduction** [7 Lectures]
   - Basic axioms of linear vector spaces and the axioms of quantum mechanics (QM)
   - Basic properties of functions and establishing the functions as vectors in infinite dimensional linear vector space, connection to bras and kets in QM
   - Sequence of functions and convergence criterion of the sequence and the concept of a metric
   - Theory of generalized functions, generally used in QM, with a special emphasize on Dirac-delta distribution
   - The Dirac notation and the reasons why it works
   - Matrix elements and expectation values

   (b) **Basic theory of integration of functions** [3 Lectures]
   - Normalization of wavefunctions in QM and Riemann integrable functions
   - Failure of Riemann integrability
   - Introduction to integration measure theory in general
   - Brief introduction to path integral measure in QM

   (c) **Brief introduction to Lebesgue measure and Lebesgue integration** [4]
   - Introduction to Lebesgue integration theory and its application
   - Properties of Lebesgue integrals and their relationship with Riemann integrals

   (d) **Hilbert space** [7 Lectures]
   - Definition of an inner product in a function space and its obvious connection with QM
   - Introduction to Hilbert spaces, in particular $L^2[\mathbb{R}]$
   - Distinction between the Banach space $L^1[\mathbb{R}]$ and $L^2[\mathbb{R}]$
   - The contents of L^p normed spaces in brief

- Norm induced by the inner products and the concept of orthonormal basis used in QM
- The concept of the dual vector space and their relationship with 'bra' vectors of Dirac
- The various basic inequalities as: Minkowski inequality, Cauchy-Schwarz inequality and Holder's inequality

(e) **Fourier series and Fourier transforms** [3 Lectures]

- Definition of a generalized Fourier series and Fourier transform and their connection with the basics of QM
- Discussion on the convergence properties of the Fourier series and Fourier transforms and their connection with Bessel's inequality and Parseval relation
- Norm preserving Fourier transforms and L^2 spaces and the relationship between the position and momentum basis in quantum mechanics
- Fourier transforms and wave packets in QM

(f) **Linear operators in function space** [7 Lectures]

- Bounded operators in function spaces
- Difficulties with unbounded operators in quantum mechanics
- The position and momentum operators as unbounded operators
- Self-adjoint bounded operators in QM
- Difficulty with self-adjoint unbounded operators and Stone-von Neumann theorem and application in basics of QM
- The Hellinger-Toeplitz theorem on symmetric operators and its huge implication in QM
- Self-adjoint extension of symmetric operators in quantum mechanics

(g) **Eigenvalue spectrum of Hermitian operators** [8 Lectures]

- The eigenvalues and eigenfunctions of linear operators in function spaces and in particular their relevance in QM
- The spectral theorem for bounded linear operators
- Some remarks on the spectral theorem for unbounded operators and their implications in QM and other branches of physics

10. **Prerequisites**: Mathematical Method I, Quantum Mechanics I

11. **Short summary for the course of study booklet**: While teaching quantum mechanics or Mathematical methods one does not get enough time to emphasize on the particular difficulties one faces when problems purely related to functional spaces arise. These problems are not at all esoteric in nature, the problems are there right in the first postulates of quantum mechanics where one assumes a fundamental commutation relation between unbounded operators. The problems again come when one has to define self-adjointness of unbounded operators in quantum mechanics. In quantum mechanics most of the simple operators as position, momentum or even the Hamiltonian turn out to be unbounded operators. To have a proper understanding of these operators and the Hilbert space on which they act, one must have to have some introductory understanding of functional space. This course is aimed to infuse some of these ideas which are very important for a physicist who wants to have a holistic grasp on quantum mechanics/quantum field theory.

12. **Recommended books**:

- 'Mathematics for Physics and Physicists' by Walter Appel
- 'Introductory Functional Analysis With Applications' by Erwin Kreyszig
- 'Principles of Quantum Mechanics' by R. Shankar
- 'Modern Quantum Mechanics' by J. J. Sakurai
- 'A Physicist's Introduction to Algebraic Structures' by Palash B. Pal

Date: 23rd May 2025



Proposer: Kaushik Bhattacharya



DPGC Convener(PHY): Sagar Chakraborty

The course is approved/not approved

Chairman, SPGC

Date: